

Gerard Bartolome

GitHub: [polyglotDataNerd](#)

Objective

Lead the design and implementation of scalable, technology-agnostic data platforms and ecosystems. I drive architecture as a lead individual contributor by shaping standards, patterns and infrastructure for analytics at scale, combining hands-on engineering with long-term platform thinking across data warehousing, streaming and cloud infrastructure.

Technical Skills

Languages: Rust; Java (8–11); Go (1.15–1.20); Scala (2.12); Python (2–3); Node.js; C#; C++; SQL dialects (T-SQL, PL/SQL, ANSI SQL); Terraform HCL; Helm (Kubernetes package manager)

Blockchain: Avalanche, Solana, EVM chains (Ethereum, Polygon, Cardano, Tron, Binance, Fantom)

ETL/ELT: Custom Golang/Scala/Java applications, Apache Spark (3.x), dbt, Apache Kafka, Apache Flink, AWS Kinesis, AWS Lambda, Talend Open Studio, IBM Datastage, Informatica, MS SSIS

Data Stores: Clickhouse, PrestoDB, Trino, Delta.io lakehouse, Amazon Redshift, PostgreSQL, MySQL, Redis, Cassandra (3.x), DynamoDB, Oracle (9i–11g), SQL Server (2008–2014)

Business Intelligence: Tableau (8–9), Looker, Apache Hue, Apache Superset, Metabase; Business Objects BI Platform 4.0, MS Reporting Services, Crystal Reports

Data Modeling: ERwin, MS Visio, Lucidchart

Software: Open-source big data stack

Systems/OS: Linux, macOS, Windows and other Unix-like environments

Speaker: AWS re:Invent 2019 – Implementing a data lake on Amazon S3

Education

Louisiana State University, Baton Rouge, LA (1997–2001)

Bachelor of Science in Computer Science / Managerial Information Science

Work Experience

Ava Labs – Lead Staff Data Engineer (Dec 2022 – Present)

Environment: AWS S3, Apache Kafka, Kubernetes, EMR, Aurora, Spark 3.4.1, Trino, Clickhouse

Languages: Go (1.20), Scala (2.12), Python, SQL, HCL Terraform, Helm

Data Platform: AWS S3/Trino/Clickhouse, S3 lakehouse, Airflow, dbt, Kubernetes

Blockchain: Avalanche C/X/P chains, Avalanche subnets, EVM-based chains

Design:

- Architected migration from Snowflake to an internal data lakehouse
- Migrated Apache Spark from AWS EMR to Kubernetes and deployed Trino on Kubernetes as the main compute engine and Clickhouse as the OLAP layer
- Applied the medallion architecture to design a cloud-based data lakehouse, integrating scalable data warehouses with dbt as the modeling layer

Developed:

- Built core infrastructure using Kubernetes, Apache Kafka, Airflow and Apache Spark for data ingestion and processing
- Developed a reusable Go worker pool to manage WebSocket interactions across multiple blockchains, streaming data into Apache Kafka
- Designed and implemented an orchestration layer to process blockchain streams via Kafka, enabling real-time analytics on the lakehouse

- Created a Spark utility library hosted in S3, providing tools for PII cleansing, Delta.io read/write operations and Spark Streaming utilities
- Built an end-to-end pipeline for Avalanche blockchain data from the data lake to Snowflake Marketplace

thredUP – Lead Staff Data Engineer (Nov 2020 – Apr 2022)

Environment: AWS (S3, Kafka, EKS, EMR, Redshift, Aurora RDS), Spark 3.1.1, Databricks, Jenkins

Languages: Java, Scala, Python, Node.js, SQL, HCL Terraform, Helm

Data Platform: AWS Redshift/PrestoDB, Looker, Delta lakehouse, Airflow, Kubernetes

Design:

- Architected the migration of the data warehouse to a data lake and lakehouse
- Served as SME for migrating the legacy MPP (Redshift) to PrestoDB on Kubernetes
- Applied the Kimball methodology to design a warehouse model in an MPP and a cloud-based data lakehouse, creating a scalable and flexible data store for analytics

Developed:

- Built ingestion microservices, containerized them with Docker and deployed to AWS EKS using Helm and Jenkins
- Developed an orchestration layer to convert Redshift data to Apache ORC for consumption by PrestoDB
- Built a Spark utility library shared via Artifactory, handling tasks from PII cleansing to Spark Streaming into the lakehouse

sweetgreen – Principal Big Data Engineer/Architect (Feb 2018 – Mar 2020)

Environment: AWS (Kinesis, Lambda, API Gateway, EMR, Redshift, RDS, SQS), Spark 2.4, PostgreSQL, MySQL, Redis, DynamoDB, Docker to AWS Fargate

Languages: Java, Go, Python, Scala/Spark, Node.js, SQL, HCL

Data Platform: AWS Redshift/EMR-Hadoop, Hadoop Hue, RESTful API services

Design:

- Applied the Kimball methodology to design a warehouse model in an MPP and a cloud-based data lake, creating a scalable and flexible data store
- Developed a CCPA framework to anonymize PII data in the sweetgreen data lake, including ad-hoc deletion from the object store upon user request

Developed:

- Built ingestion microservices, containerized them and deployed to AWS Fargate using Terraform
- Built open-source ETL/ELT applications that integrate disparate sources into a language-agnostic data platform on AWS
- Consolidated data sources for user experience, event-driven streams, analytics and personalization across store operations and digital customers
- Built a review scraper using Go's goroutines and channels to fetch public reviews for ML sentiment analysis
- Built a serverless clickstream interface to ingest real-time event payloads into Spark and Redshift for CRM and analytics
- Built an optimization application to predict store ingredient demand within 15-minute intervals based on sales, forecasts, labor and customer traffic

Beachbody On Demand – Big Data Engineer/Architect (Oct 2015 – Feb 2017)

Environment: AWS (Kinesis, Lambda, API Gateway, EMR, Redshift, RDS, SQS), Hadoop, Spark, PostgreSQL, MySQL, Cassandra 3.0, DynamoDB

Languages: Java, Python, Node.js, HiveQL, Scala/Spark, SQL

Data Platform: AWS Redshift/EMR-Hadoop/Kinesis

Design:

- Applied the Kimball methodology to create a warehouse model and a cloud-based data lake to build and integrate data into a scalable store

Developed:

- Developed open-source ETL/ELT applications that integrate disparate sources into an AWS platform
- Built a terabyte-sized data platform integrating data from multiple sources for user experience event bus and personalization
- Built a serverless clickstream application to ingest real-time data payloads into Hadoop and Redshift for real-time recommendations and analytics
- Built a machine-learning personalization engine using Spark ML collaborative filtering based on user workout behavior
- Built an Alexa skill to capture voice recognition metrics for content usage using a serverless stack, ingesting into Redshift and DynamoDB for state

NBCUniversal/Fandango – Data Engineer/Architect (Aug 2013 – Sep 2015)

Environment: AWS, Hadoop, Spark, Omniture, PostgreSQL, MySQL, MS SQL Server

Data Platform: AWS Redshift, Tableau Server 9.0

Design:

- Applied the Kimball methodology to create a star schema model and integrate data into a scalable, schema-agnostic star-schema-like structure
- Integrated on-premises data centers with cloud environments to build a hybrid data lake

Developed:

- Utilized a variety of tools (Java, Python, Scala, Hive, Spark, Pig, procedural languages) along with cloud technologies to implement and monitor the full ETL/ELT lifecycle
- Built a terabyte-sized data platform from diverse sources for insights, machine learning, forecasting and predictive modeling

Stylesight Inc. – Data Warehouse/Business Intelligence Architect (Feb 2012 – 2014)

Environment: MS SQL Server 2012 BI Edition, MySQL 4.0–5.1, Salesforce, Omniture, Marketo

Data Platform: SAP BusinessObjects 4.0 Edge

Design:

- Applied the Kimball methodology to design a star schema and integrate data from disparate sources

Developed:

- Used SQL 2012 stored procedures as the main ETL tool along with Talend Open Studio for Salesforce extraction and SSIS for Omniture
- Utilized indexing and SQL Server 2012 optimization techniques to improve ETL performance
- Served as main DBA for SQL Server 2012, ensuring database health and query optimization
- Developed enhancements to the data warehouse and BI platform
- Built universes for clickstream, sales, client services, market strategy and finance metrics
- Created reports using BusinessObjects tools (Desktop Intelligence, Web Intelligence, Explorer)
- Performed BO administration, setup, and managed user roles and privileges

MTV Networks – Data Warehouse Architect/Developer (Jan 2010 – 2012)

Environment: Windows/Linux, MS SQL Server 2000–2008, Oracle 10g

Data Platform: BusinessObjects XI R2, BI Platform 4.0 Enterprise

Design:

- Applied the Kimball methodology to design a star schema for Forecast, Budget, Royalty, Cash and Invoice reporting

Developed:

- Used SQL 2008 stored procedures as the main ETL tool from diverse sources
- Utilized indexing and SQL Server 2008 features to optimize ETL performance
- Developed BusinessObjects universes and reports for Consumer Products and Ancillary Sales
- Provided BO administration, setup and maintenance, including roles and Active Directory privileges
- Served as lead database developer for the Consumer Products content management system, maintaining logic, developing stored procedures/triggers/functions and optimizing the database

Additional Experience

- Full career history and detailed project list available on request.